

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 15



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Recap of Previous Lecture



Topic

Mensuration 2D & 3D



Topics to be Covered



Topic-1

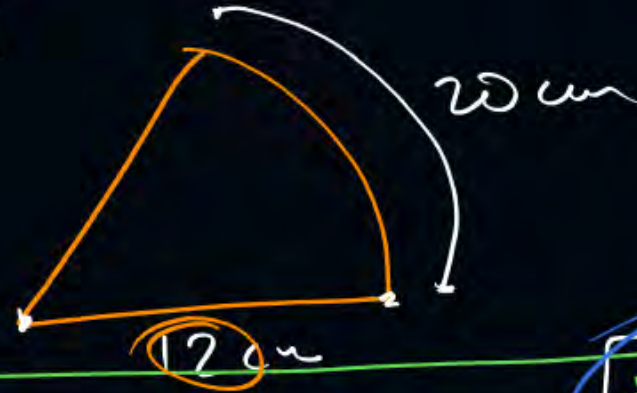
Data Interpretation



[MCQ]

#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

$A = ?$



Assignment

$$\text{Area of Sector} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Length of Arc} = \frac{\theta}{360} \times 2\pi r$$

$$\Rightarrow \frac{20}{12} = \frac{\theta}{360} \times 2\pi \times 12$$

$$\Rightarrow 120 = \frac{\theta}{360} \times \pi r^2$$

A 60 sq. cm

B 240 sq. cm

C 120 sq. cm

D 64 sq. cm

[MCQ]



#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.

Assignment

A 32 m

B 8 m

C 16 m

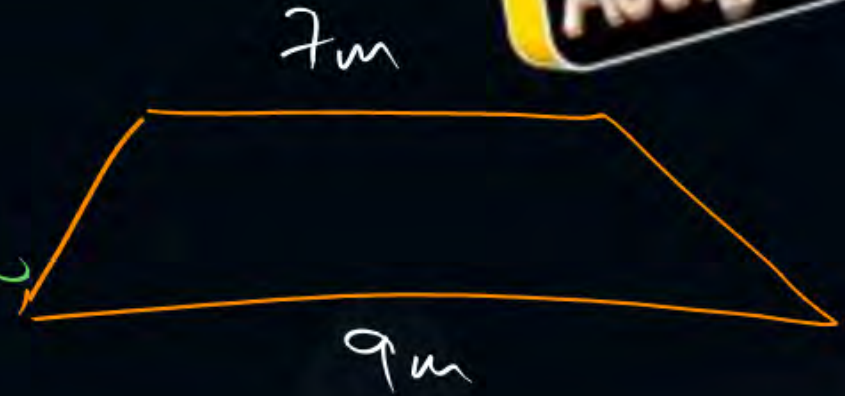
D 4 m

Area of Tresp.

$$= \frac{1}{2} \times (\text{sum of || sides}) \times h$$

$$\Rightarrow \frac{16}{128} = \frac{1}{2} \times \frac{16}{8} \times h$$

$$\therefore h = 16$$



[MCQ]



$$V = a^3$$

#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured.
Find the number of smaller cubes so formed.

Assignment

$$\text{Volume of Big Cube} = \text{No.} \times \text{Volume of smaller cube}$$

A 12

B 27

C 24

D 8

$$\Rightarrow 216 = x \times 8$$

$$\therefore x = \frac{216}{8} = \underline{\underline{27}}$$

[MCQ]

$$l = 4x \quad | \quad b = 3x \quad | \quad h = 2x$$



#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m³. Find its total surface area.

Assignment

$$\text{Volume of a cuboid} = l \times b \times h$$

A 1300 m²

B 1500 m²

C 1333 m²

D 2700 m²

$$\Rightarrow 3000 = 4x \times 3x \times 2x$$

$$\Rightarrow 3000 = 24x^3$$

$$\Rightarrow \frac{3000}{24} = x^3$$

$$x = \sqrt[3]{125} = 5$$

$$\begin{aligned} \text{ TSA} &= 2(lb + bh + hl) \\ &= 2x(300 + 150 + 200) \\ &= 2x650 = 1300 \text{ m}^2 \end{aligned}$$

[MCQ]



#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

Assignment

$$LSA = 2h(l+b)$$

$$= 2 \times 10(35)$$

$$= 20 \times 35 = \underline{\underline{700 \text{ m}^2}}$$

$$\text{Cost} = 7 \times 15 = \underline{\underline{105}}$$

A

₹ 70

B

₹ 105

C

₹ 225

D

None of these

[MCQ]



#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?

A 13 cm

B 41 cm

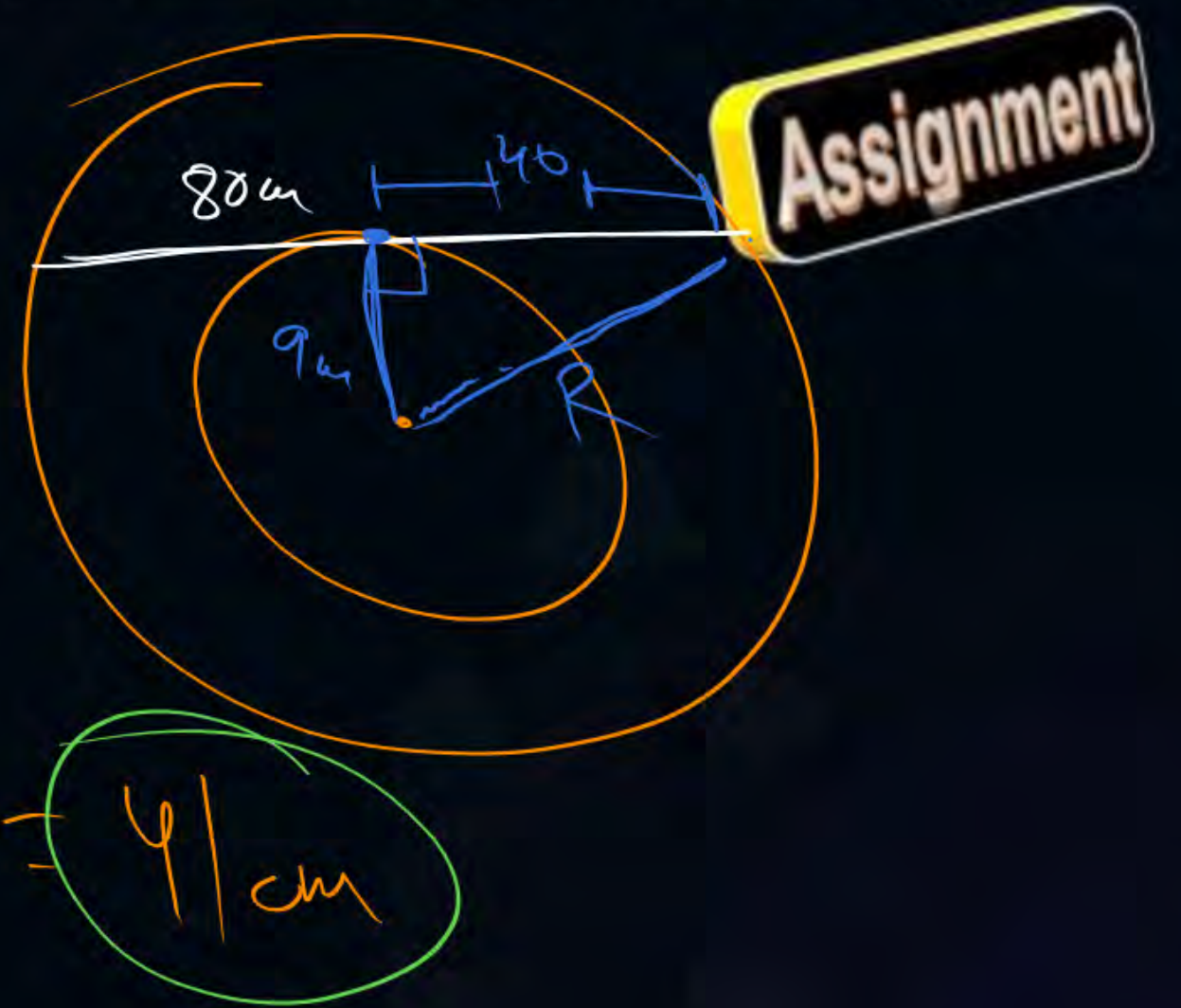
C 52 cm

D 75 cm

$$R^2 = 9^2 + 40^2$$

$$\Rightarrow R^2 = 81 + 1600$$

$$\therefore R = \sqrt{1681} = 41 \text{ cm}$$



[MCQ]

$$P = \underline{2(l+b)} = 2 \times 9 = \underline{18 \text{ cm}}$$



#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

$$(a+b)^2 = a^2 + b^2 + \underline{2ab}$$
$$\Rightarrow a^2 + b^2 = (a+b)^2 - 2ab$$

Area of rectangle = $l \times b$

$$\Rightarrow \underline{20} = \underline{lb}$$

$$l^2 + b^2 = (\sqrt{41})^2$$

$$\underline{l^2 + b^2 = 41}$$

$$(l+b)^2 - 2lb = 41$$

$$\Rightarrow (l+b)^2 - 2 \times 20 = 41$$
$$(l+b)^2 = 81$$



$$\therefore \underline{l+b=9}$$

Assignment

A

9 cm

B

18 cm

C

20 cm

D

41 cm

Types:

The numerical data pertaining to any event can be presented by any one or more of the following methods.

- 1) Tables ✓
- 2) Line Graphs ✓
- 3) Bar Graphs or Bar Charts ✓
- 4) Pie Charts or Circle Graphs ✓

Line Graph:

Study the following graph carefully and answer the questions:



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]



#Q. What is the percentage of marks scored by the student in the unit test of August and October taken together?

- A** 50%
- B** 83.3%
- C** 60%
- D** 75.5%

$$\frac{755}{1000} \times 100$$
$$= \frac{755}{10} = \underline{\underline{75.5\%}}$$



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]



#Q. What are the average of marks obtained by the student in the unit test of given academic year?

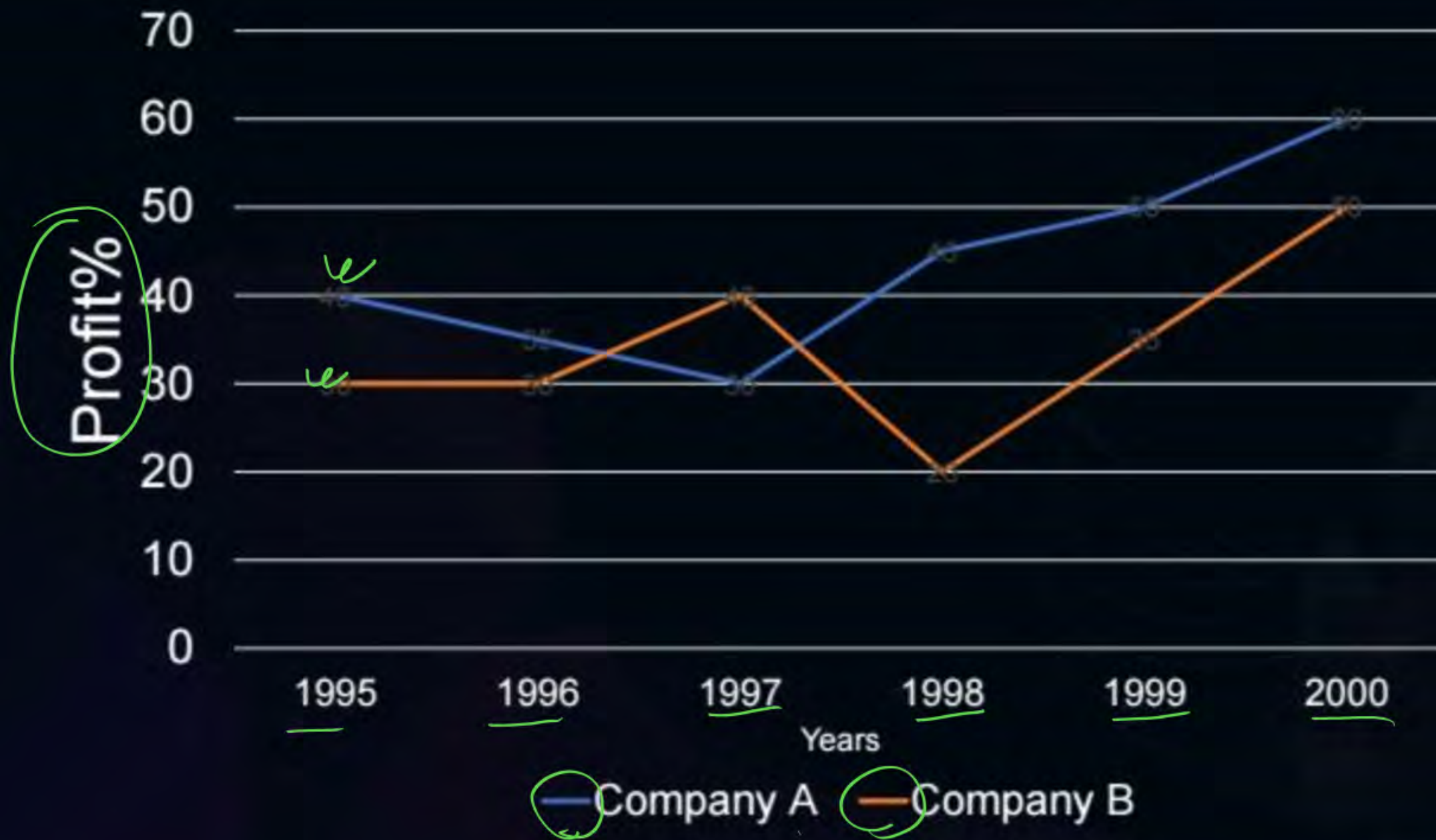
- A** 361.6
- B** 380.8
- C** 377.5
- D** 400.0

$$\frac{725 + 755 + 805}{6} = \frac{2285}{6} = 380.8$$



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

Study the following table and answer the questions:



[MCQ]

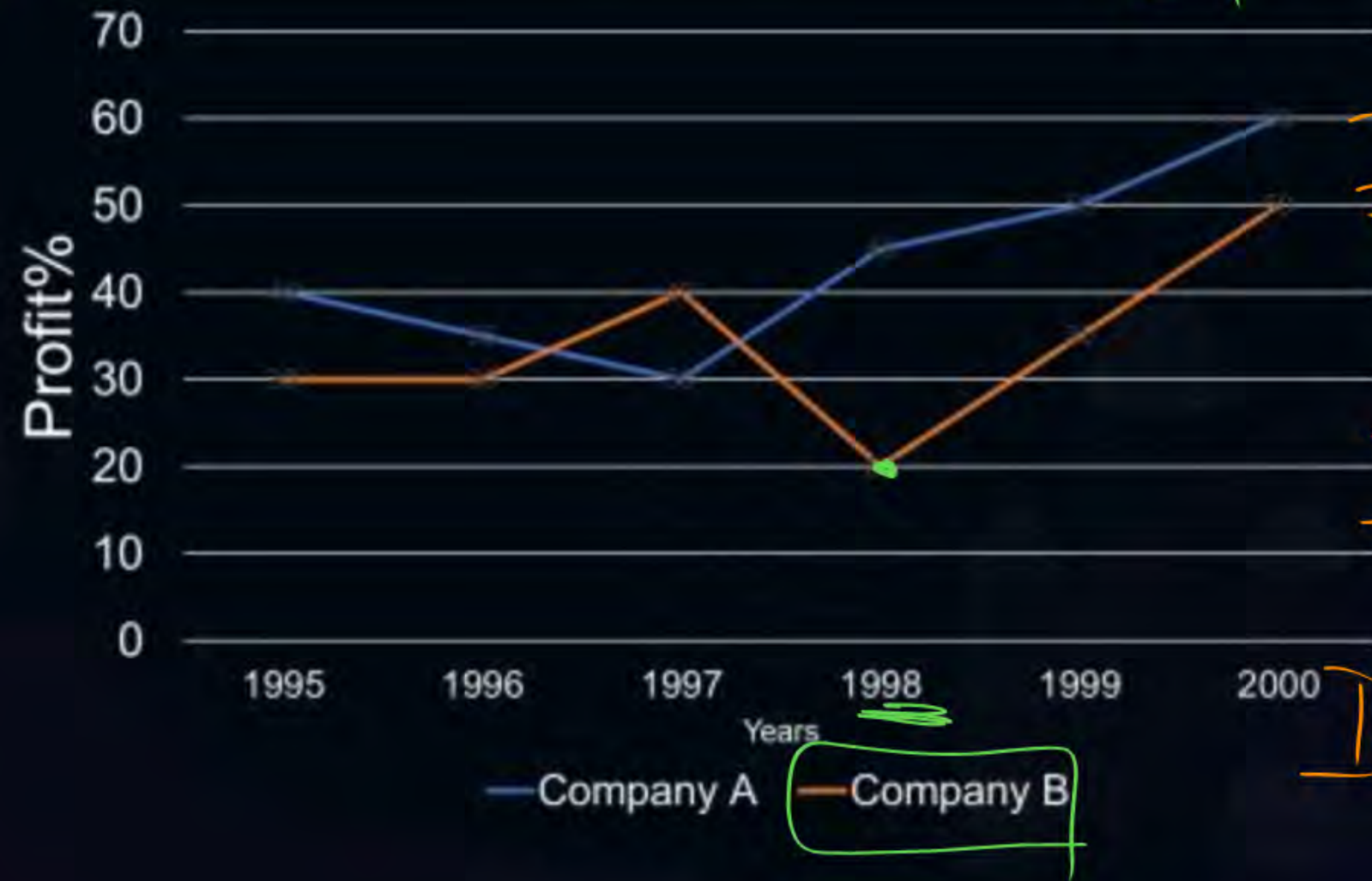


#Q. If the expenditure of the company B in the year 1998 was ₹200 crore, what was its income?

$$P\% = 20\%$$

$$\frac{S.P}{C.P} = 1.2$$

- A 360 crores
- B 300 crores
- C 240 crores**
- D 120 crores



$$\frac{\text{Income}}{\text{Exp}} = 1.2$$

$$\text{Income} = 1.2 \times 200$$

$$\text{Income} = \underline{\underline{240}}$$

[MCQ]



#Q. If the income of company A in the year 2000 was ₹600 crore, what was its expenditure?

$$P\% = 60\%$$

$$\Rightarrow \frac{\text{Income}}{\text{Exp}} = 1.6 \Rightarrow \frac{600}{\text{Exp}} = 1.6$$

A

210 crores

B

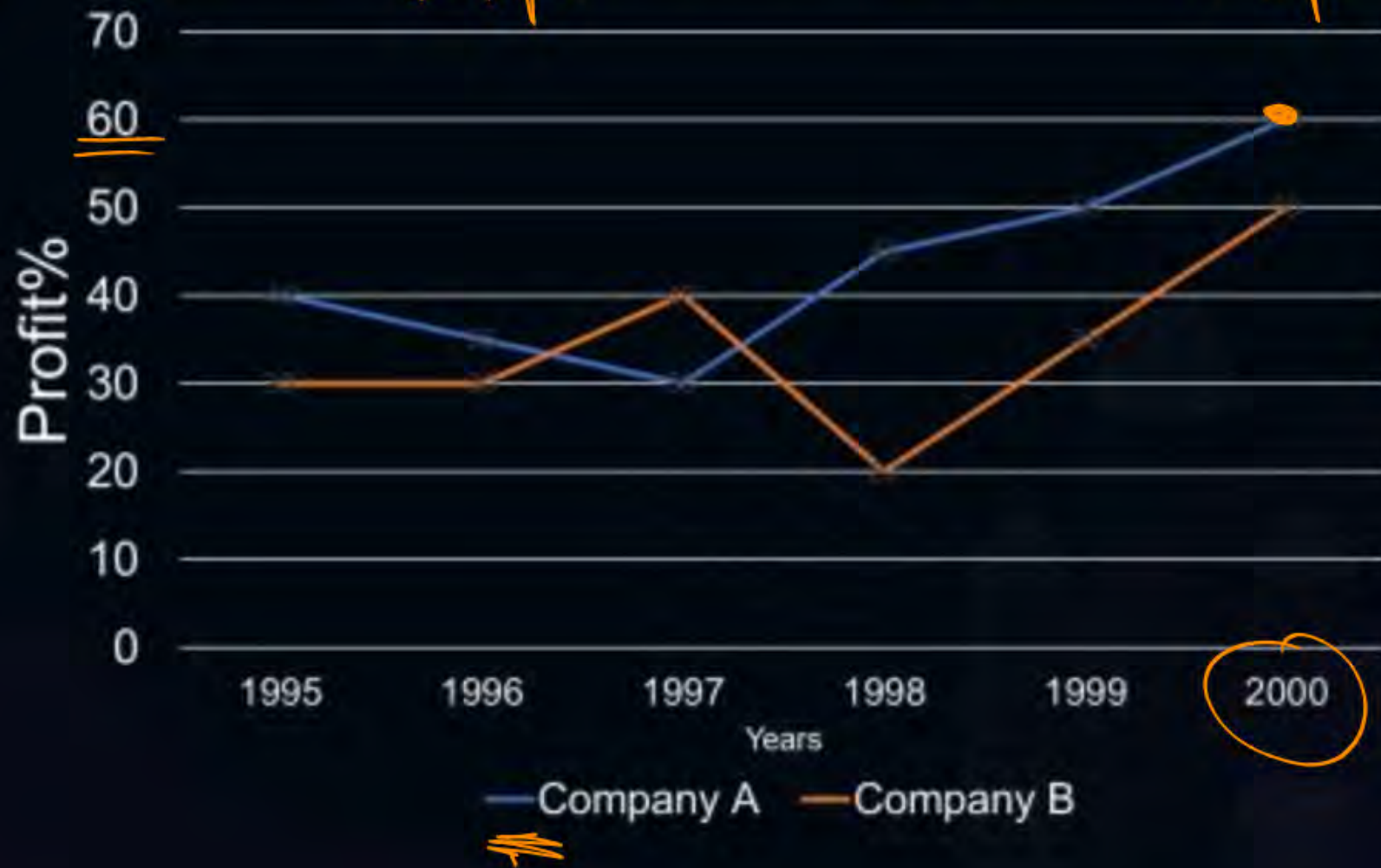
275 crores

C

340 crores

D

375 crores



$$\text{Exp} = \frac{600}{1.6} = 375$$

[MCQ]



$$\underline{I_A = I_B}$$

#Q. If the income of the two companies were equal in 1996, what was the ratio of their expenditure?

$$\underline{E_A : E_B = 26 : 27}$$

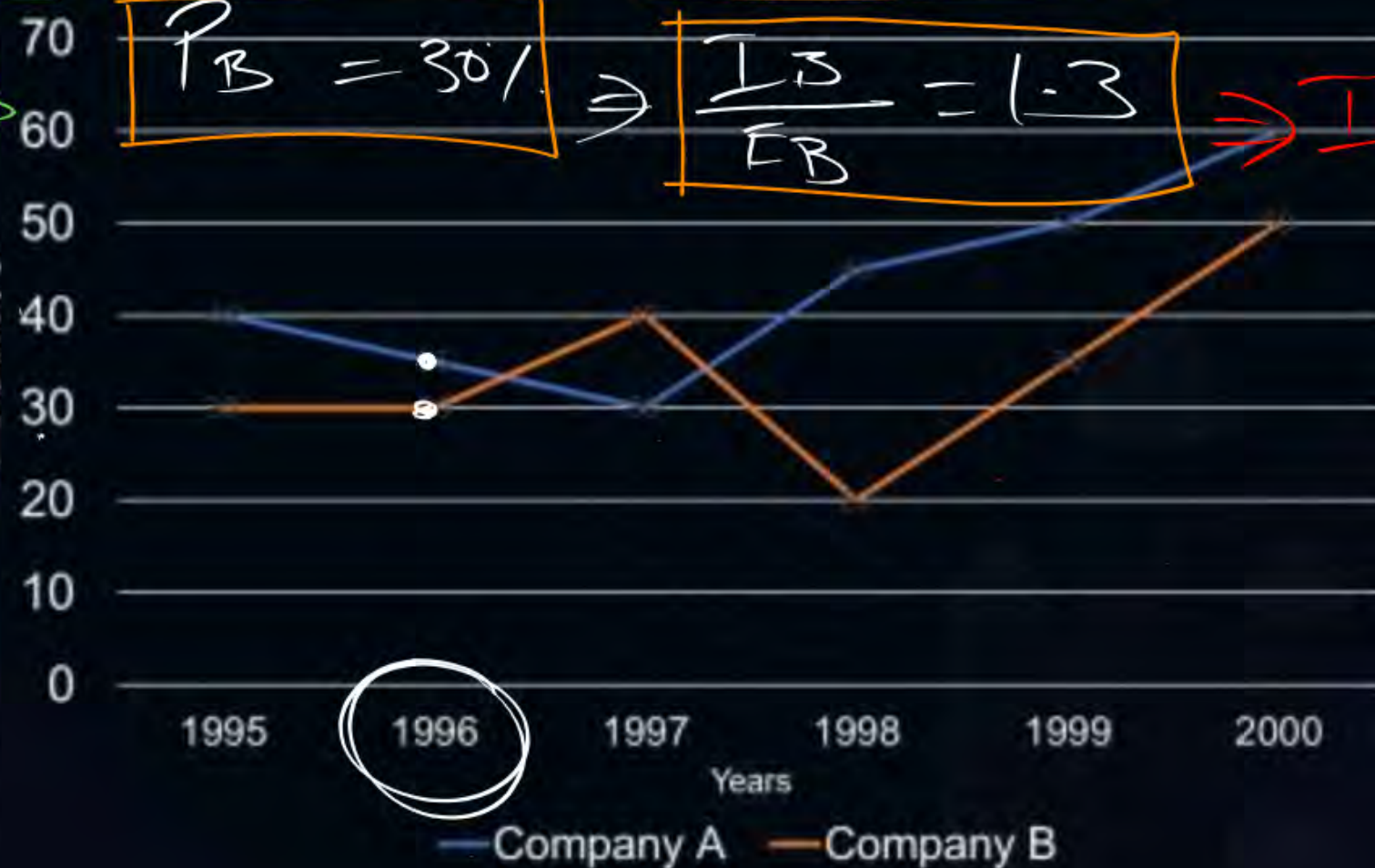
$$1.35 E_A = 1.3 E_B$$

$$\frac{E_A}{E_B} = \frac{1.3}{1.35}$$

$$= \frac{\cancel{135} 26}{\cancel{135} 27}$$

$$P_A = 35\% \Rightarrow \frac{I_A}{E_A} = 1.35 \Rightarrow I_A = 1.35 E_A$$

$$P_B = 30\% \Rightarrow \frac{I_B}{E_B} = 1.3 \Rightarrow I_B = 1.3 E_B$$



A

26 : 27

B

100 : 67

C

1 : 2

D

5 : 27

The following graph depicts the number of students who joined and left the school in the beginning of the academic year.

Strength of school in 2005 was 3000
2005 में स्कूल की संख्या 3000 थी

Assignment



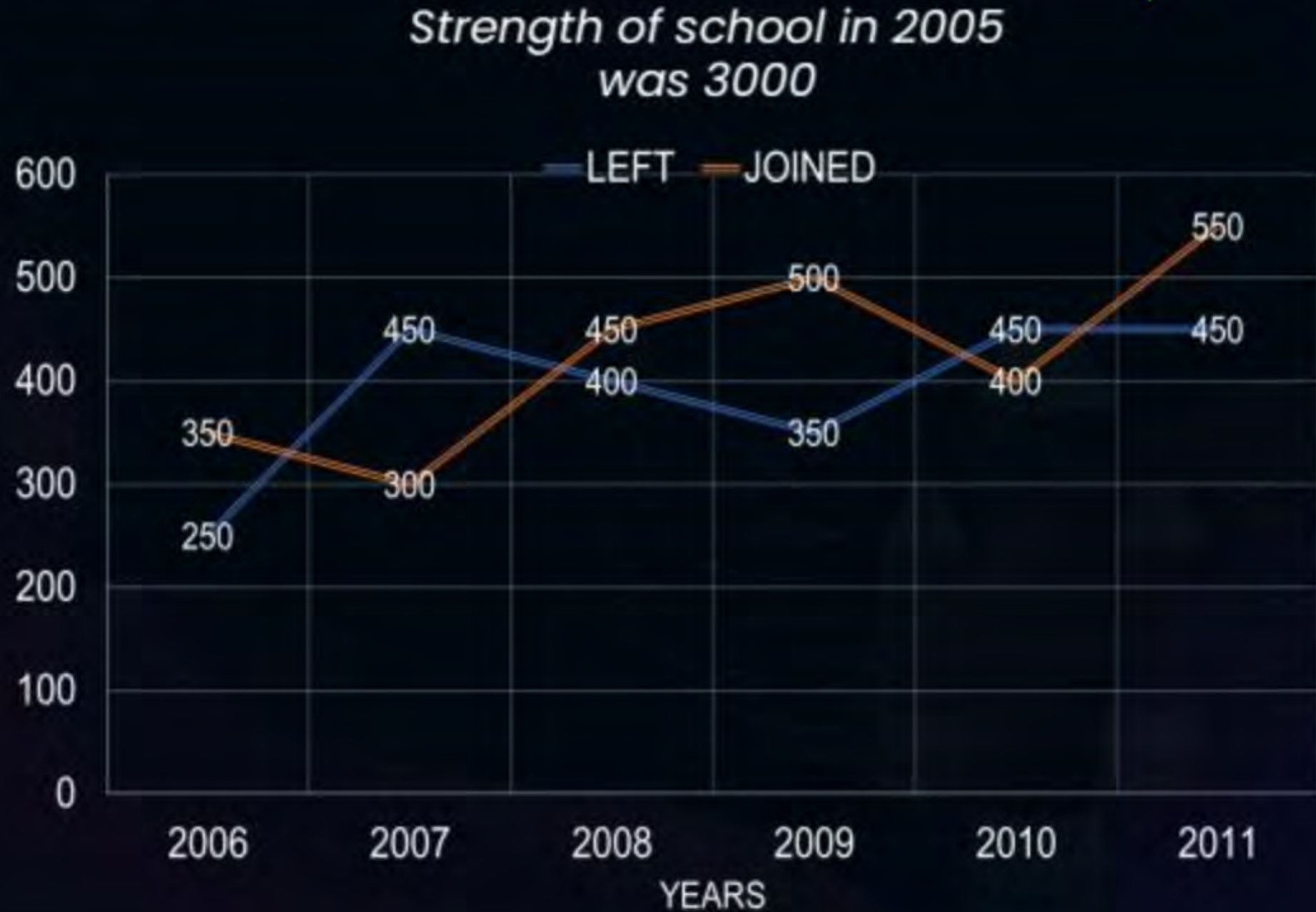
[MCQ]



#Q. By approximately what percent, the strength of school increased or decreased from 2007 to 2008?

Assignment

- A** 1.2%
- B** 1.7%
- C** 2.1%
- D** 2.4%



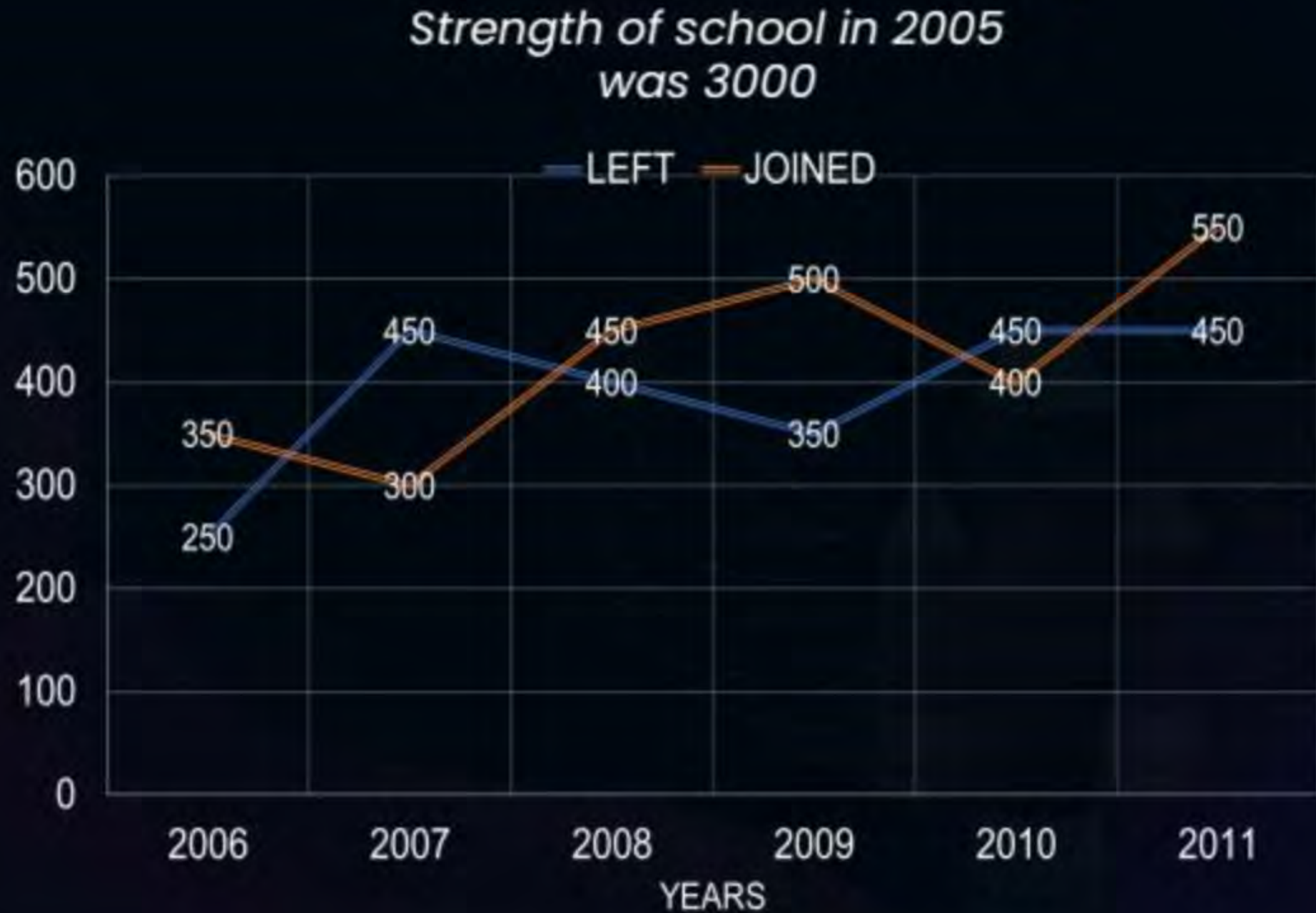
[MCQ]



#Q. How many students were studying in the school during 2009?

Assignment

- A** 2950
- B** 3000
- C** 3100
- D** 3150



[MCQ]



#Q. The number of students studying in school in 2008 was what percent of the number of students studying in the school in 2011?

Assignment

Strength of school in 2005
was 3000

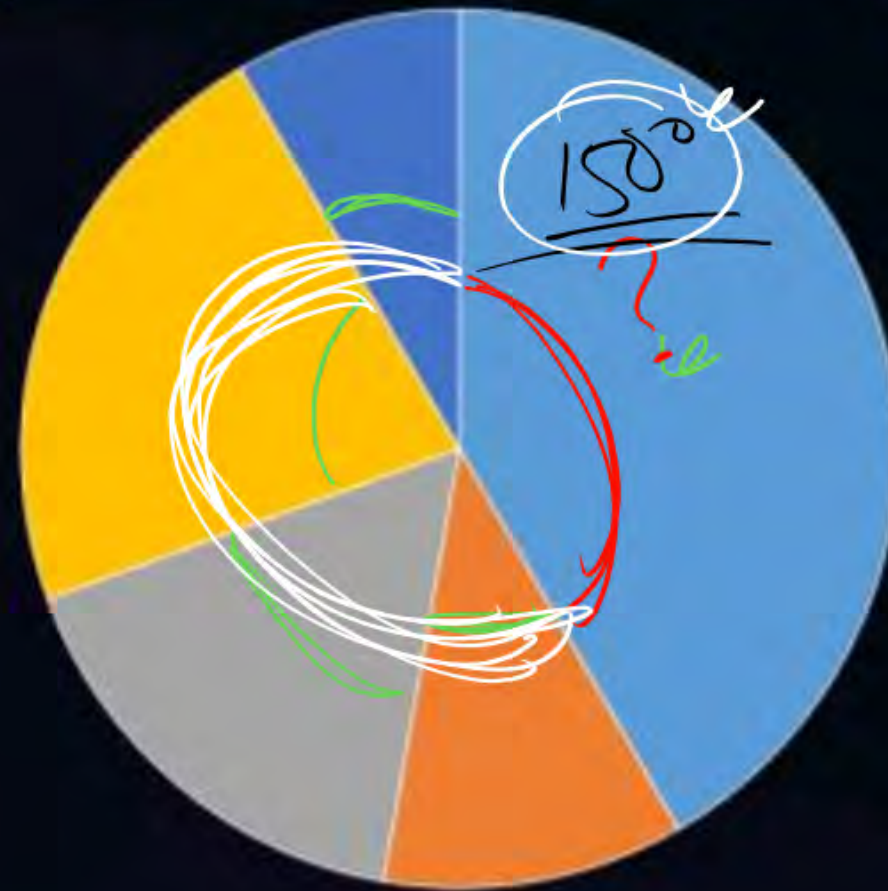
- A** 92.13%
- B** 93.75%
- C** 97.25%
- D** 96.88%



Pie Chart:

Study the following graph carefully and answer the questions:

POPULATION OF 24000



■ UNEMPLOYED

■ SELF EMPLOYED - 40° ✓

■ PUBLIC SECTOR - 60° ✓

■ PRIVATE SECTOR - 80° ✓

■ GOVT. SECTOR - 30° ✓

$$360 - 210 = 150^\circ$$

40

[MCQ]

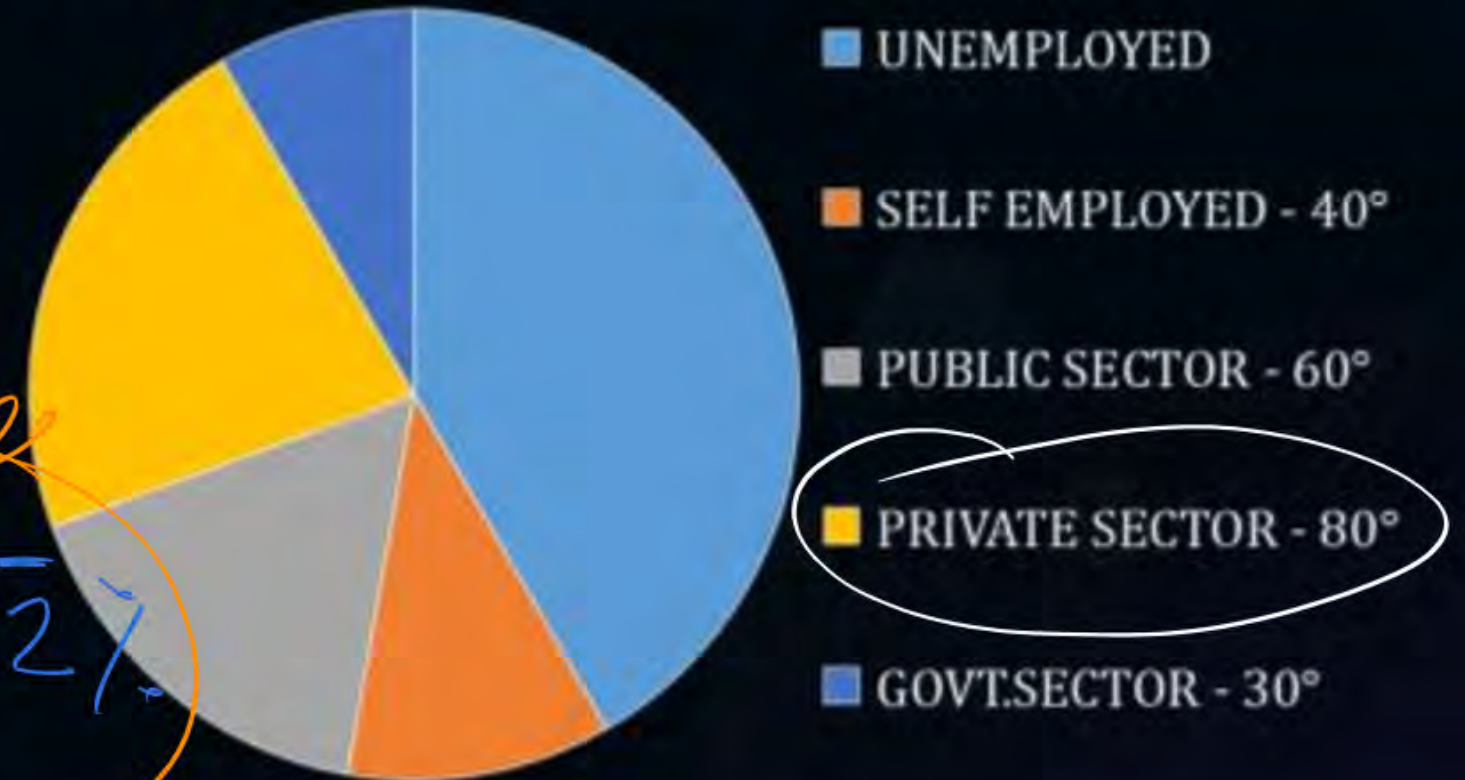


#Q. If private sector employees are to be mentioned in ^{below} ~~above~~ chart in the form of percentage, it would be%.

$$\frac{20}{80} \times 100$$
$$\frac{20}{80} \times 100$$
$$= 25$$

$$= \frac{25}{9} = 22.2\%$$

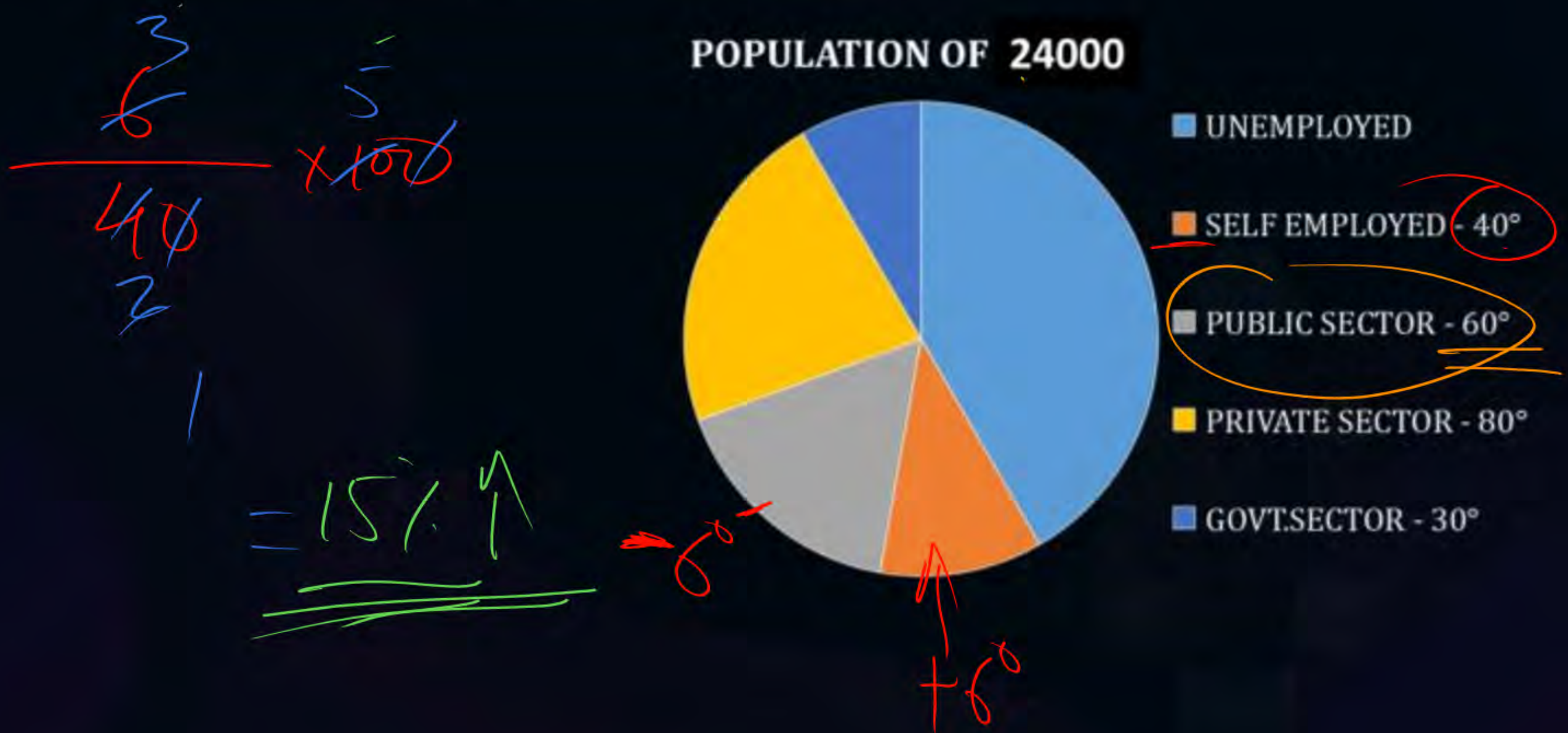
POPULATION OF 24000



[MCQ]



#Q. If 10% of public sector people resign and becomes self-employed, find the increase percent in self-employed population.



[MCQ]

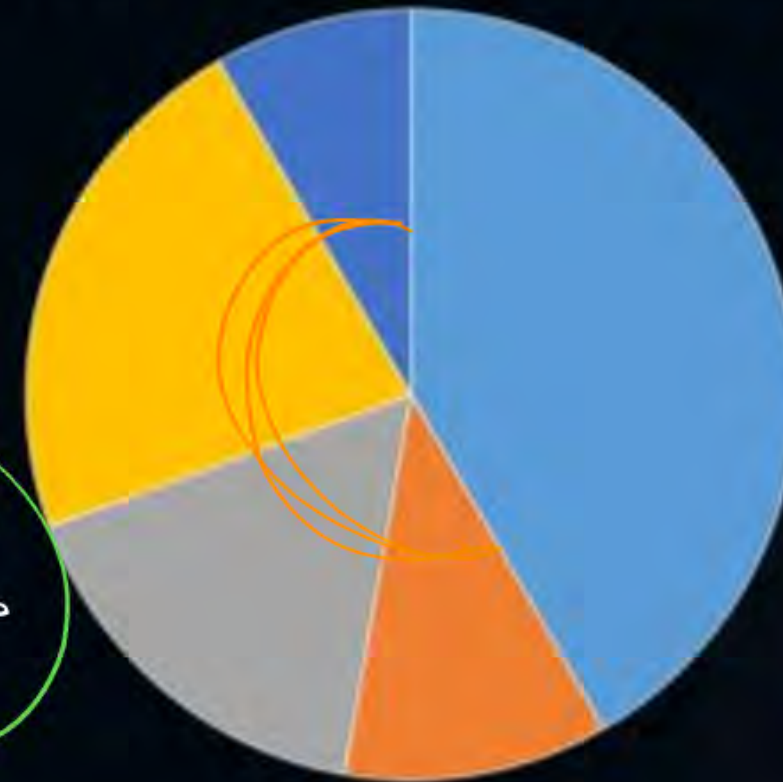


#Q. What percentage of employed people are working in government sector?

$$\frac{30}{210} \times 100$$

$$= \frac{100}{7} = 14.28\%$$

POPULATION OF 24000



UNEMPLOYED

SELF EMPLOYED - 40°

PUBLIC SECTOR - 60°

PRIVATE SECTOR - 80°

GOVT.SECTOR - 30°

210

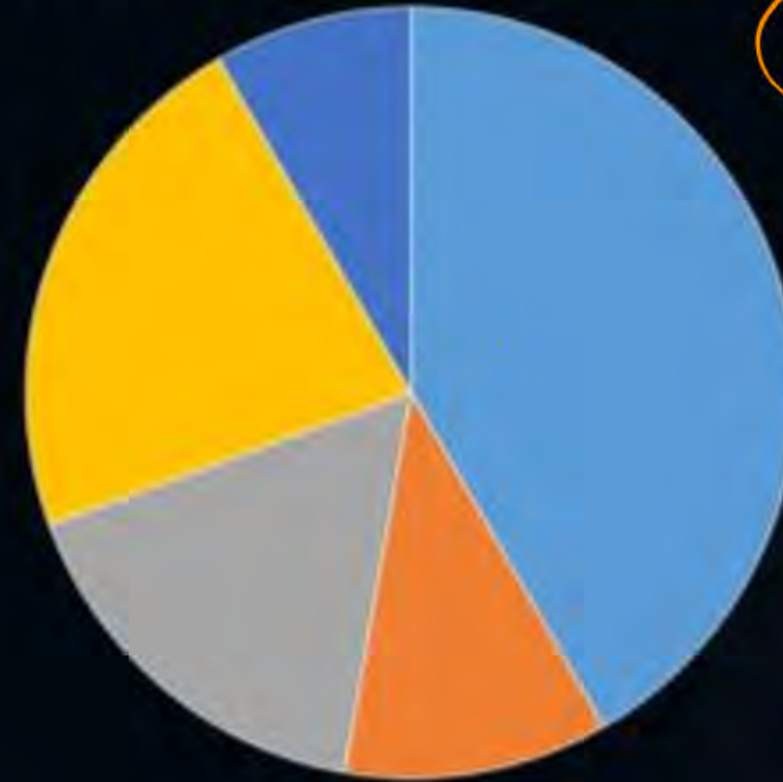
[MCQ]



#Q. Find the number of people who are unemployed.

$$\begin{array}{r} 5 \\ 150 \\ \hline 360 \\ 12 \\ 1 \end{array} \quad \begin{array}{r} 2500 \\ \times 24000 \\ \hline 10,000 \end{array}$$

POPULATION OF 24000



UNEMPLOYED

SELF EMPLOYED - 40°

PUBLIC SECTOR - 60°

PRIVATE SECTOR - 80°

GOVT. SECTOR - 30°

$$360 - 210 = 150^\circ$$

Tabular form:

Study the following table and answer the questions:

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]



#Q. Who among the following employees earns maximum bonus in comparison to his total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A

P

B

Q

C

R

D

S

[MCQ]



#Q. The income from overtime is what percent of income from arrears in the case of employee P?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A

90

B

80

C

40

D

75

[MCQ]



#Q. How many employee have their salary more than 3 times the income from bonus?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

- A** One
- B** Five
- C** Three
- D** None

[MCQ]



#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A

T

B

P

C

S

D

R

[MCQ]



#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A

P

B

Q

C

T

D

S



2 mins Summary



Topic

Data Interpretation



THANK - YOU